

2023



Modernizing Legacy Applications Using AWS AML with Asure

Learn how Asure Software transformed its HCM application and reduced legacy Microsoft licensing dependencies using AWS AML.

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30,000 employees' payroll processed | **9.2% reduction**

in <4 minutes



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Improves

system availability

Modernized

legacy Microsoft Windows Server stack



3 AWS Certifications earned

as part of AWS AML

Overview

As a provider of human capital management (HCM) solutions, [Asure Software](#) (Asure) has continually evolved and adopted new technological advancements. Although the company had migrated to Amazon Web Services (AWS), its legacy applications were still running on Microsoft Windows Servers and were not optimized for running in the cloud. Payroll processing is a sensitive business, and Asure needed to modernize its applications so that it could offer optimal availability and reliability.

This process required not only technical expertise but also resources to upskill employees and drive a “true cloud modernization” mindset. To support this journey, Asure engaged [AWS Application Modernization Lab](#) (AWS AML), a program that helps remove Microsoft license dependencies and accelerate innovation with open-source and cloud-based technologies. Thus, the company transformed its HCM solutions, boosting agility and reducing costs by nullifying long-term Microsoft license commitments.



Opportunity | Using AWS AML to Optimize HCM Solutions for Asure

Founded in 1985, Asure provides cloud-based HCM solutions for small- and medium-size companies. Its payroll and human resources solutions have been adopted by 100,000 businesses in the United States, encompassing 1.7 million employees.



“People want to be paid on time, and it is a big deal when they’re not,” says John Canada, vice president of engineering at Asure. “In addition to being able to scale, we wanted our systems to have extremely high availability. So, we aimed to modernize our legacy Microsoft Windows Server technology stack, reduce our cost and dependencies on Microsoft licensing, and increase uptime as much as possible for our customers.”

In 2017, Asure performed a lift-and-shift migration to AWS and ran its legacy applications in the cloud using self-managed [Amazon Elastic Compute Cloud](#) (Amazon EC2) for Microsoft Windows Server, which provides secure and resizable compute capacity. However, the company remained burdened by Microsoft licensing costs and commitments. Asure chose to work alongside AWS AML to refactor its software and reduce the Microsoft licensing burdens for Windows Server, SQL Server, and .NET Framework, thus facilitating a true modernization journey on AWS.

“Working alongside the AWS team, we can accelerate our modernization strategy by updating legacy systems, nullifying the need for Microsoft licensing, and rearchitecting our solution,” says Yasmine Rodriguez, chief technology officer of Asure. “This not only optimizes our cloud environment but also empowers us to respond quickly to HCM market demands and changes.”

Solution | Processing Payroll for 30,000 Employees in Less than 4 Minutes

Collaborating with AWS AML, Asure gained access to the expertise and resources that it needed to rearchitect its legacy Microsoft Windows Server technology stack, teach its employees cloud best practices, and deliver much-needed training and certification opportunities. As part of AWS AML, Asure engaged [AWS Professional Services](#), a global team of experts that helps companies realize their desired business outcomes on AWS, for hands-on technical support.

“Alongside the AWS team, we built an outstanding solution for our use case, reduced our Microsoft licensing burdens, and reduced our costs by 9.2 percent. We estimate total cost savings of 30 percent upon completion,” says Canada. “The team walked us through the modernization of our current applications, on both the backend and the front end, and continues to support us as we build new services.”



To adapt to the new environment, Asure’s engineers needed to learn new skills. The company engaged [AWS Training and Certification](#), which equips teams with practical cloud skills that increase return on investment and accelerate business results, to level up its team.

As part of its collaboration with AWS AML, Asure earned three [AWS Certifications](#), which validate technical skills and cloud expertise, and participated in three AWS Immersion Day events, which are designed to educate engineers about AWS services. These events focused on technologies that were key to application modernization, such as [AWS Lambda](#), a serverless, event-driven compute service. The company went on to earn a total of eight AWS Certifications and schedule three more AWS Immersion Day events after the engagement.

“By upskilling the engineers, we’ve improved the retention of developers,” says Rodriguez. “They’re excited about what they’re learning and how they’re growing. They see that we’re investing in them. Our collaboration with AWS AML has been instrumental in this whole journey.”

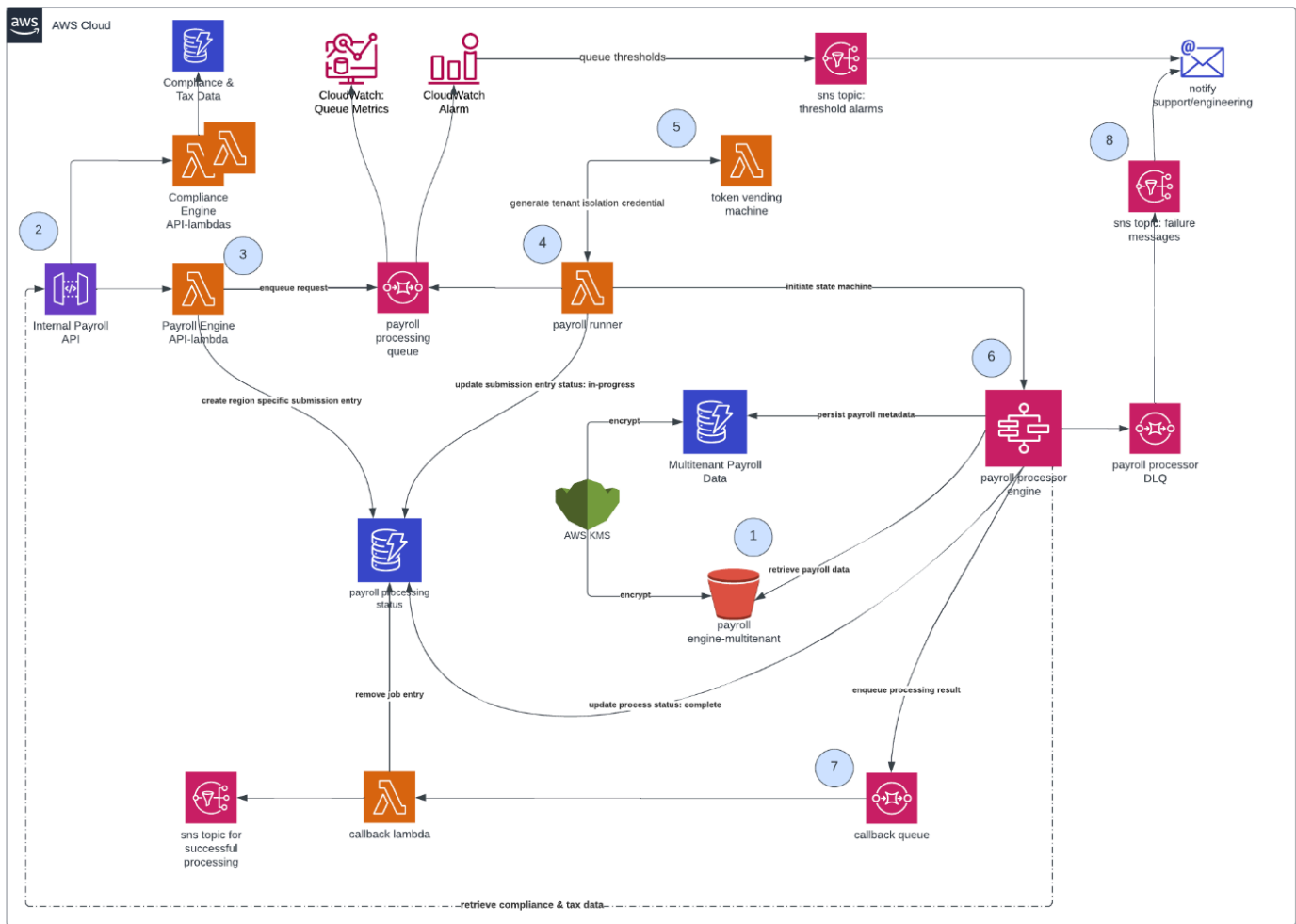
Receiving this training, Asure built an event-driven, asynchronous payroll engine on AWS (see figure 1. Payroll engine architecture). After clients submit payroll information, API calls managed by [Amazon API Gateway](#)—a service that is used to create, maintain, and secure APIs at nearly any scale—initiate payroll processing. AWS Lambda is used to facilitate communication with the API, load jobs into the queue, and process the payroll logic.

The processed paychecks are stored in [Amazon Simple Storage Service](#) (Amazon S3), an object storage service built to retrieve virtually any amount of data from anywhere. Asure also created an active/active disaster recovery strategy to enhance its resilience and availability while maintaining optimal uptime for its customers.

Customers’ data is stored in [Amazon DynamoDB](#)—a fast, flexible NoSQL database. To keep this data secure and give only authorized users access to it, Asure created a token vending machine using [AWS Identity and Access Management](#) (AWS IAM), which is used to securely manage identities and access to AWS services. The company also uses AWS IAM to implement tenant isolation.

With this modernized system, Asure needs less time to process payroll. For example, it can process payroll for 30,000 employees in less than 4 minutes. “Processing payroll for 2,000 employees would have taken a long time in our legacy system,” says Canada. “On AWS, payroll processing is 80 times faster; payroll runs for thousands of employees reduced from minutes to seconds. The performance gains have been remarkable.”

Architecture Diagram



Outcome | Continuing to Modernize HCM Software to Keep Pace with Innovation

Asure’s collaboration with AWS AML has ushered in a new era of innovation, efficiency, and resilience. The company plans to roll out its newly architected payroll engine to production by the end of 2023 and will soon work alongside AWS Professional Services to modernize the engine’s user interface.

“It’s one thing to work with a vendor that essentially tells you what to do,” says Andrew Quartey, principal architect at Asure, “but it’s quite another to work with one that invests in your success. The AWS team is on this journey with us.”

About Asure Software

Asure Software helps business owners grow their teams while minimizing risk and overhead by reducing legacy Microsoft licensing burdens for applications and databases, such as Windows Server, SQL Server, and .NET Framework.

AWS Services Used



AWS Application Modernization Lab (AWS AML)

Accelerate innovation with open source and cloud-native technologies

[Learn more »](#)

AWS Professional Services

Helping you achieve your desired business outcomes with AWS

[Learn more »](#)

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

Amazon API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

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